

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2008

BIOLOGY
Higher 2

9747/03

PAPER 3 Applications Paper
25 August 2008

1 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and candidate number in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Answer **all** questions.

The intended number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English.

FOR EXAMINER'S USE	
1	
2	
3	
4	
TOTAL	65

Answer **all** the questions in spaces provided.

1. Genetically modified plasmids were the first vectors developed and are still widely used for cloning of genes. Such plasmids are often used to introduce foreign genes into bacterial cells and these recombinant bacteria will in turn be used either in the mass production of useful proteins or in the transformation of cells to form genetically modified organisms.

(a) What are plasmids?

_____ [1]

(b) State **three** essential features of a plasmid for it to be used as a vector for the cloning of genes in bacteria.

_____ [3]

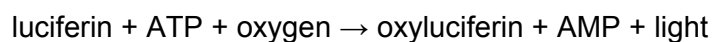
(c) Using a **named** example, explain the advantages of being able to transfer human genes into bacteria.

_____ [3]

(d) Suggest why plasmids by themselves cannot be used to insert genes directly into plant cells.

_____ [1]

There is a concern that transgenic organisms pose a threat should they be released into the environment. One method of detecting such organisms is to introduce a further gene that codes for an enzyme, **luciferase**, along with the desired gene. This enzyme occurs naturally in a number of marine bacteria and causes bioluminescence (the production of light) in the presence of oxygen and the substrate luciferin. The reaction that takes place is shown below.



(e) What is meant by a **transgenic organism**?

_____ [2]

(f) Describe the main steps involved in producing colonies of transformed *E. coli* with the luciferase gene.

_____ [4]

(g) Suggest an advantage of using the luciferase gene to detect the presence of escaped genetically modified organisms. [1]

[Total:15]

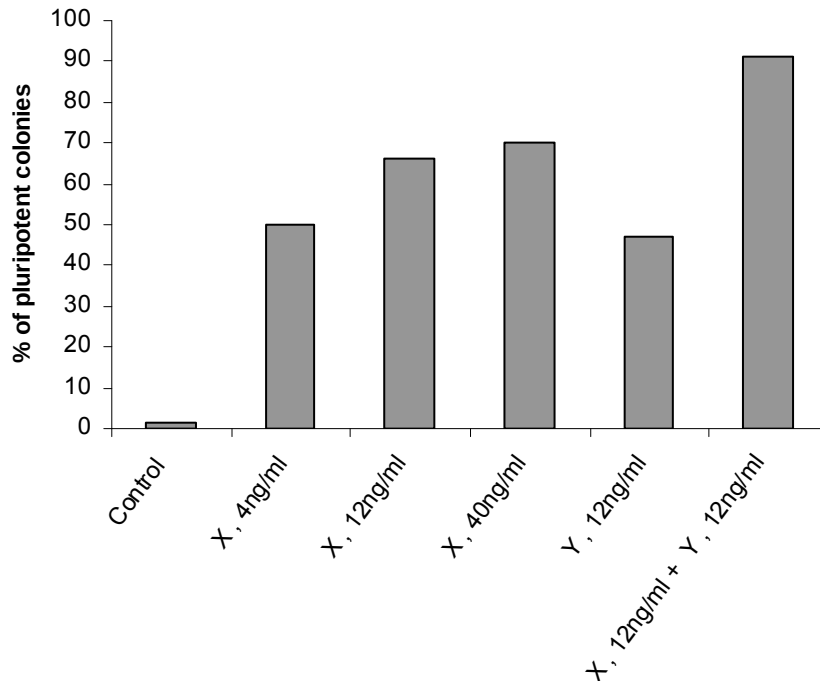
2. In order to study embryonic stem cells, scientists extract the inner cell mass from blastocysts and culture the cells *in vitro*. The embryonic stem cells must be grown in a special culture medium that maintains the pluripotency status of the cells. Without the appropriate growth factors, the embryonic stem cells will spontaneously differentiate.

(a) Telomerase is not usually active in cells of multicellular organisms. With reference to a feature of stem cells, suggest whether embryonic stem cells possess active telomerase.

[2]

Two growth factors, **X** and **Y**, were added to embryonic stem cell culture to study their effect on the pluripotency status of the stem cells. (**Fig. 2.1**).

Fig. 2.1 Effect of growth factors X and Y on the pluripotency of cultured embryonic stem cells



- (b) (i) With reference to Fig. 2.1, describe the effects of **X** and **Y** on the pluripotency of cultured embryonic stem cells.

[3]

- (ii) Both growth factors are protein molecules which are not able to enter the embryonic stem cell. Suggest how the growth factors are still able to cause the cells to remain pluripotent.

[2]

- (c) The use of embryonic stem cells is ethically controversial as embryos are destroyed in the process. As such, scientists have devised another way of obtaining pluripotent stem cells.

Differentiated mouse skin cells were 'reprogrammed' by infection with retroviruses carrying the genes for four different transcription factors. With the incorporation of these four genes, the cells became pluripotent and showed many characteristics of embryonic stem cells. These cells were termed induced pluripotent stem (iPS) cells.

Fig.2.2 below shows how scientists have cured mice with sickle-cell anaemia. iPS cells were generated from a sick mouse (steps 1-4). The sickle-cell allele in the iPS cells was replaced using gene therapy (steps 5-6). The iPS cells with the functional allele were made to differentiate into blood stem cells (step 7) before transplantation back into the same mouse (step 8), which cured the mouse (step 9).

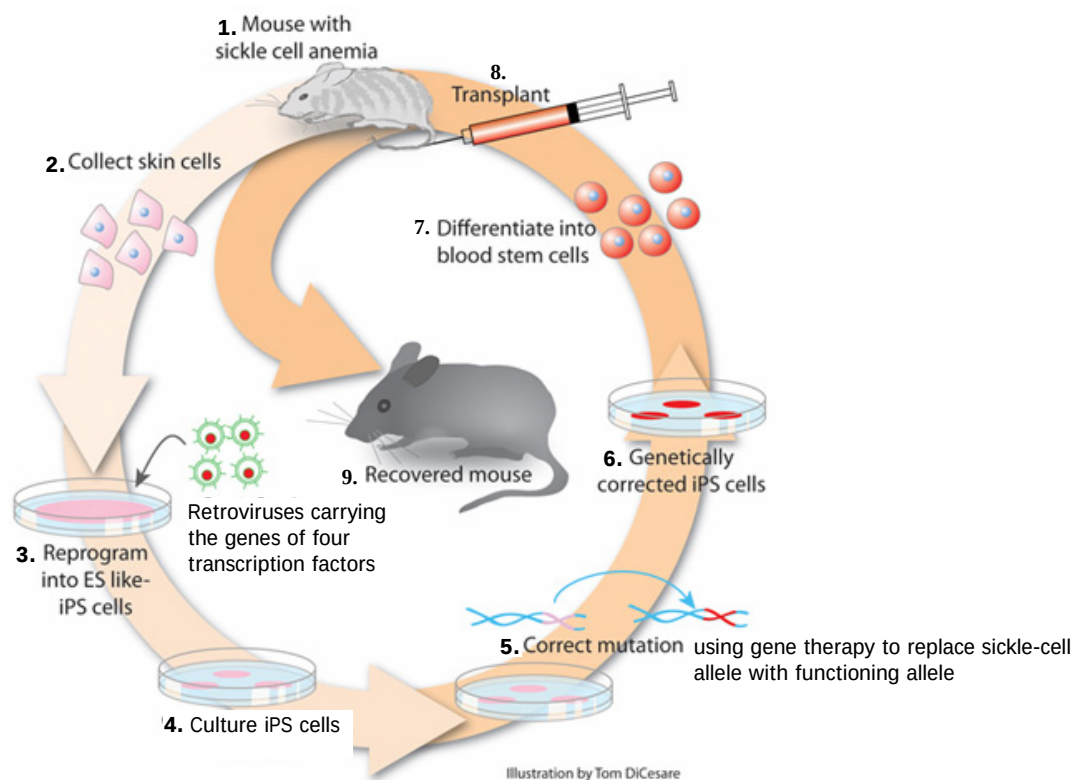


Fig. 2.2

- (i) State the function of transcription factors.

[1]

- (ii) Suggest how the insertion of genes coding for transcription factors can cause a differentiated cell to become pluripotent.

[2]

- (iii) Describe **three** social and ethical considerations for the use of gene therapy.

[3]

- (d) Unlike animals, all cells of a plant are totipotent. This characteristic has been exploited in producing genetically-modified crops as one genetically-modified plant cell is able to divide and differentiate to form a whole plant.

Describe, with named examples, **three** improvements in quality and yield of crop plants that have been achieved using genetic engineering.

[3]

[Total: 16]

3. Figure 3.1 below shows the process of Southern blotting using a sample.

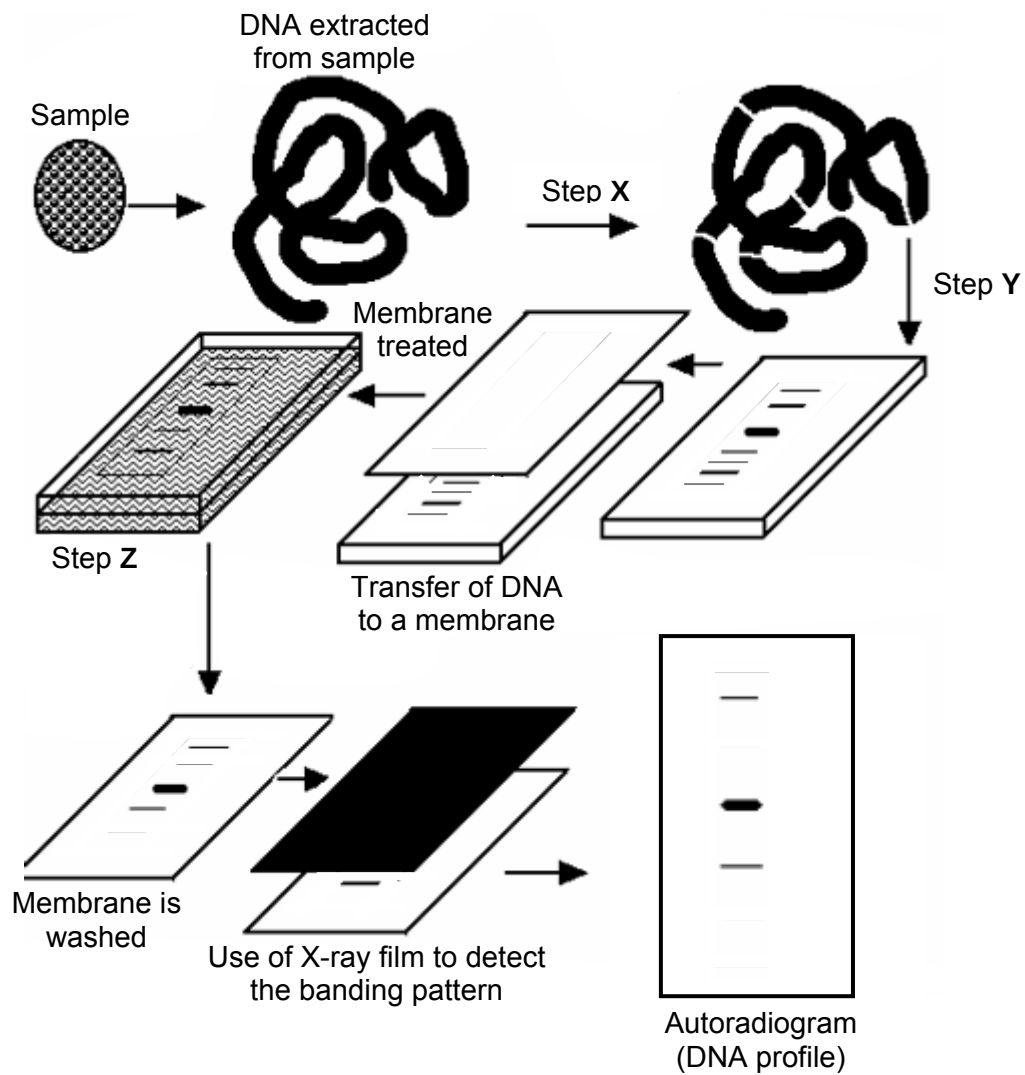


Fig. 3.1

a) i) Briefly explain the following steps:

Step X

[2]

Step Y

[2]

Step Z

[2]

- (ii) Suggest a reason for the need to produce the autoradiogram for analysis rather than using the results obtained after Step Y.

[1]

Figure 3.2 shows the DNA profiles of two transgenic plants and a couple of bacteria that are found in the soil around these plants.

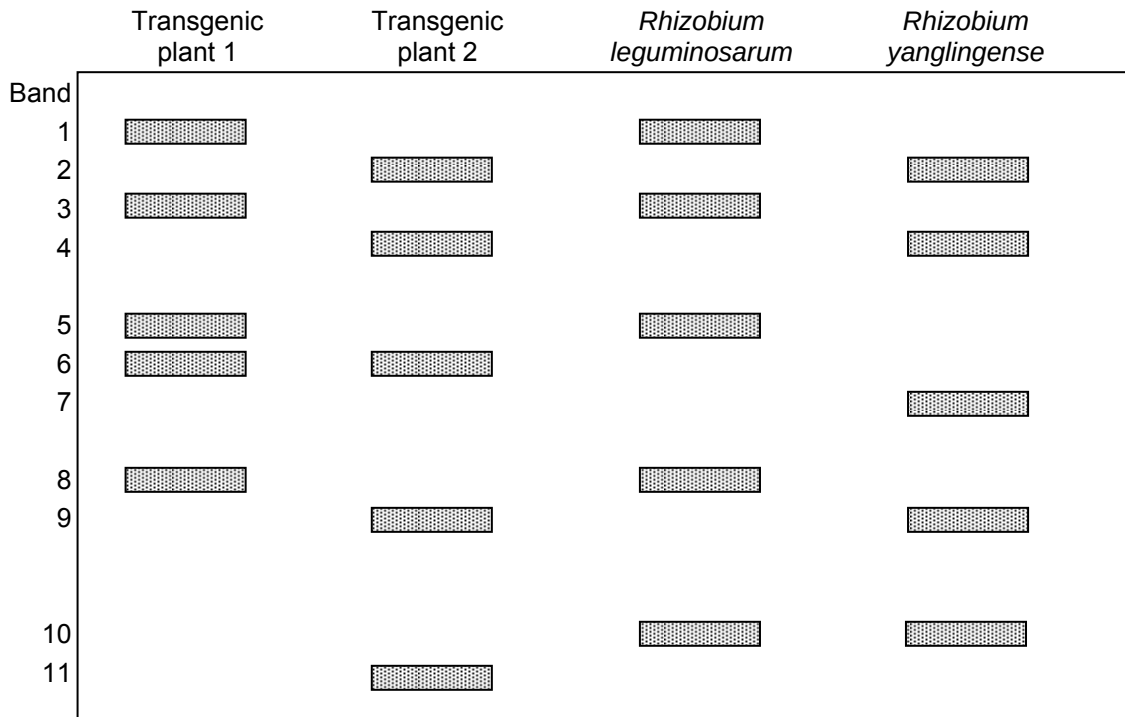


Fig 3.2

- (b) (i) Describe the process by which bacteria are able to obtain DNA from their surroundings.

[1]

- (ii) Based on **Fig 3.2**, match the bacteria to the transgenic plant that it is likely to be found with. Explain your answer.

[3]

- (iii) Suggest **one** implication of growing genetically modified plants shown by the DNA profiles in Fig 3.2.

[1]

- (iv) Suggest why the quality of transgenic plants grown from seeds produced by existing transgenic plants may differ from those grown by tissue culture.

[2]

[Total: 14]

Write your answers to this question on the ruled pages provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a)** Explain how RFLP analysis facilitates the process of

- (i) disease detection [4]
- (ii) genomic mapping [2]

- (b)** What are the benefits that can be derived from sequencing the human genome? [7]

- (c)** Discuss briefly the ethical issues raised by the Human Genome Project. [7]

Handwriting practice lines consisting of 20 sets of three horizontal lines (top, middle, and bottom) for letter formation.

Blank lined area for writing.

